VU Advanced Software Engineering **Sheet 1: Abstract Interpretation**

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Exercise 1: Widening

Perform Abstract Interpretation with the standard interval domain as in the lecture on function `verify_with_widening` below. After each `goto` instruction perform widening as in the lecture. Provide the abstract control flow graph, whose edges need to be labeled either with the corresponding instruction, with `JOIN` or with `WIDEN`. The last state of your graph should correspond to line 9. What does the analysis reveal about the safety of `verify_with_widening`?

```
 0  void verify_with_widening(int x) {
 1      if (x >= 0)
 2          x = -1;
 3      int i = x;
 4      if(i < 1){
 5          i = i + 2;
 6          x = x * 2;
 7          goto 4;
 8       }
 9      assert(x < i);
10  }
```

Exercise 2: Modified Interval Domain

Consider the modified interval domain INT_d with the following set of abstract elements: $\{[a, b] \mid a \geq 0 \text{ and } 0 \leq b \leq 2 + a\} \cup \{\bot, [0, \infty]\}$. INT_d cannot represent negative numbers which is why we restrict ourselves here to programs over unsigned integers. Perform Abstract Interpretation with INT_d on function `check_verified` below. Provide the abstract control flow graph as in the lecture. What does the analysis reveal about the safety of the assertion? Comparing INT_d with the standard interval domain, does the modification of the interval domain come with precision loss? Justify your answer.

```
0  void check_verified(unsigned int y) {
1      unsigned int x = 3;
2      unsigned int z = x;
3      if(y <= 1){
4          x = y + 1;
5          z = x * x;
6      }
7      assert(x != 0 && z < 5);
8  }
```

Exercise 3: Abstract Interpretation without Widening

Consider again the function Foo from the lecture:

```
0  void Foo(int i) {
1       int x = 5;
2       int y = 7;
3       if (i >= 0) {
4            y = y + 1;
5            i = i - 1;
6            goto 3;
7       }
8       assert(0 <= y - x);
9  }
```

Give a modified interval domain INT' s.t. an Abstract Interpreter using INT' verifies Foo without using widening. Give the abstract control flow graph of the Abstract Interpretation run like in the lecture. What is the height of your abstract domain INT'?

After that give a new program P containing an assertion that cannot be verified with an Abstract Interpreter that uses INT' but can be verified with an Abstract Interpreter using the standard interval domain from the lecture. Again, you are not allowed to use widening.